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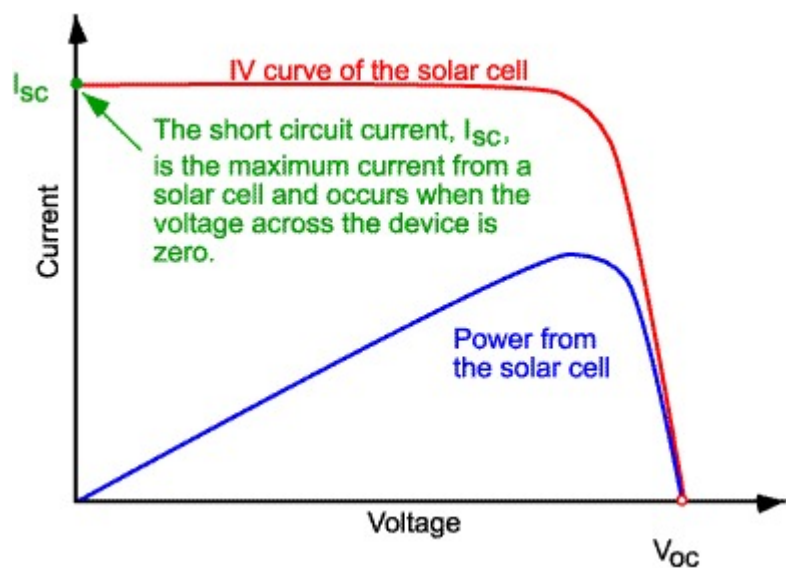
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Short-Circuit Current

The short-circuit current is the current through the solar cell when the voltage across the solar cell is zero (i.e., when the solar cell is short circuited). Usually written as I_{SC} , the short-circuit current is shown on the IV curve below.



IV curve of a solar cell showing the short-circuit current.

The short-circuit current is due to the generation and collection of light-generated carriers. For an ideal solar cell at most moderate resistive loss mechanisms, the short-circuit current and the light-generated current are identical. Therefore, the short-circuit current is the largest current which may be drawn from the solar cell.

The short-circuit current depends on a number of factors which are described below:

- * **the area of the solar cell.** To remove the dependence of the solar cell area, it is more common to list the short-circuit current **density** (J_{sc} in mA/cm^2) rather than the short-circuit current;
- * **the number of photons** (i.e., the power of the incident light source). I_{sc} from a solar cell is directly dependant on the light

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Korean Version

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intensity as discussed in [Effect of Light Intensity](#) ;

- * **the spectrum of the incident light.** For most solar cell measurement, the spectrum is standardised to the **AM1.5 spectrum** ;
- * **the optical properties** (absorption and reflection) of the solar cell (discussed in [Optical Losses](#)); and
- * **the minority-carrier collection probability** of the solar cell, which depends chiefly on the surface passivation and the minority carrier lifetime in the base.

When comparing solar cells of the same material type, the most critical material parameter is the diffusion length and surface passivation. In a cell with perfectly passivated surface and uniform generation, the equation for the short-circuit current density can be approximated as:

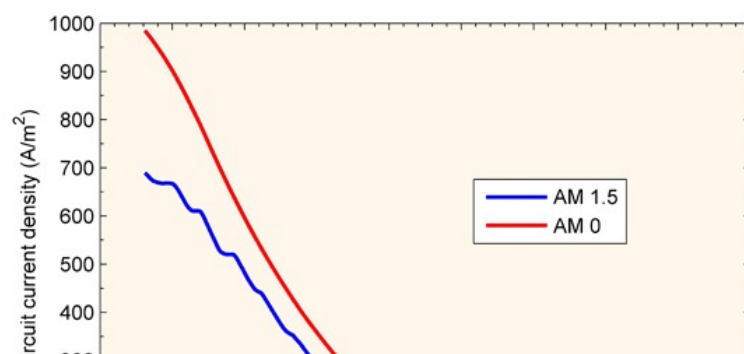
$$J_{SC} = qG(L_n + L_p)$$

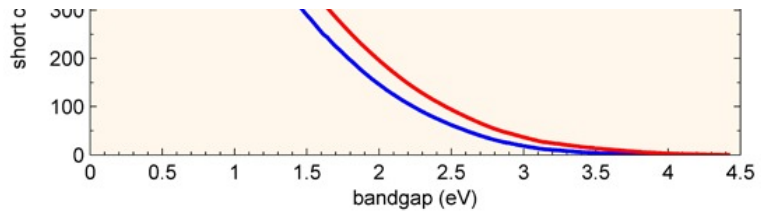
where G is the generation rate, and L_n and L_p are the electron and hole diffusion lengths respectively. Although this equation makes several assumptions which are not true for the conditions encountered in most solar cells, the above equation nevertheless indicates that the short-circuit current depends strongly on the generation rate and the diffusion length.

The short circuit current, I_{SC} , is the short circuit current density, J_{SC} , times the cell area:

$$I_{SC} = J_{SC}A$$

Silicon solar cells under an AM1.5 spectrum have a maximum possible current of 46 mA/cm². Laboratory devices have measured short-circuit currents of over 42 mA/cm², and commercial solar cell have short-circuit currents between about 28 mA/cm² and 35 mA/cm².





In an ideal device every photon above the bandgap gives one charge carrier in the external circuit so the highest current is for the lowest bandgap.

Illuminated Current and Short Circuit Current (I_L or I_{SC} ?)

I_L is the light generated current inside the solar cell and is the correct term to use in the solar cell equation. At short circuit conditions the externally measured current is I_{SC} . Since I_{SC} is usually equal to I_L , the two are used interchangeably and for simplicity and the solar cell equation is written with I_{SC} in place of I_L . In the case of very high series resistance ($> 10 \Omega\text{cm}^2$) I_{SC} is less than I_L and writing the solar cell equation with I_{SC} is incorrect.

Another assumption is that the illumination current I_L is solely dependent on the incoming light and is independent of voltage across the cell. However, I_L varies with voltage in the case of drift-field solar cells and where carrier lifetime is a function of injection level such as defected multicrystalline materials.

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